Learning Unified Policies for Adaptive Grasp Recovery

Multi-Skill Imitation Learning and Reinforcement Learning for Impact-Induced Grasp Recovery

Li Jiaqi

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Part I: Revised Proposal (3 slides) · Part II: Milestone Report (3 slides)

Proposal 1/3 Problem and Research Question

Problem

A robot successfully grasps and lifts a cylindrical object. However, an incoming ball may strike the object and destabilize the grasp, causing:

- Slipping or displacement within the gripper
- Unwanted rotation or partial loss of contact
- Complete detachment or eventual dropping

Different post-impact states require different recovery behaviors.
Unnecessary correction may also disturb a grasp that remains stable after impact.

Research Question

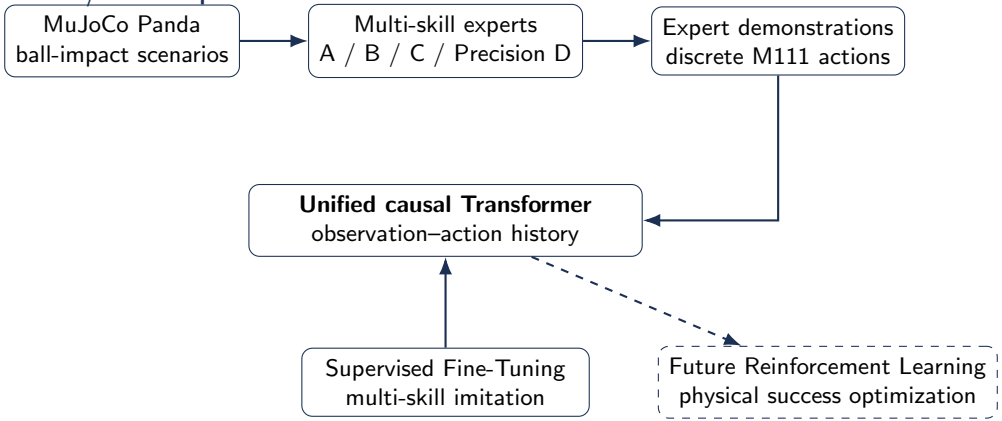
Can a single learned policy use observation history to select and execute appropriate recovery behaviors for different post-impact grasp states?

Platform:
MuJoCo simulation of a Franka Panda robot with a two-finger gripper, manipulating a cylindrical object subjected to ball impacts.

Goal

Recover grasps destabilized by external impacts while avoiding unnecessary corrections to stable grasps.

Proposal 2/3 Updated Method



Policy Interface

$$\pi_\theta(a_t \mid o_{0:t}, a_{0:t-1}), \qquad a_t \in \{\text{M111 control actions}\}$$

Proposal 3/3 Evaluation Plan

Controller Comparisons

- Nominal controller without recovery
- Skill-specific expert controllers
- Individual-skill imitation policies
- Unified imitation policy
- RL-refined unified policy (future)

Primary Evaluation

Physical Recovery Success

After recovery control ends, evaluate whether the object remains successfully grasped during a 12-second physical stability test.

Evaluate different recovery behaviors:

- A/B: recovery from disturbed grasps
- C: recatching after contact loss
- D: precise pose correction

Additional Metrics

- Object dropping, slipping, and contact loss
- Recovery success across different impact conditions
- Generalization to unseen grasp states

Research sequence: Expert demonstrations → imitation learning → unified policy → reinforcement learning.

Milestone 1/3 What Has Been Built

Simulation and Recovery Framework

- MuJoCo robotic grasping environment with external ball impacts
- Multiple expert recovery behaviors for different post-impact states
- Reproducible simulations and physical recovery evaluation

Learning Pipeline

- Expert demonstrations converted into discrete actions
- Causal Transformer trained through supervised imitation learning
- Closed-loop evaluation of learned recovery policies

Why Discrete Actions?

Different experts perform different recovery skills. A shared discrete action vocabulary allows one policy to learn from all of their demonstrations.

Current progress: Expert demonstrations and the imitation-learning pipeline have been implemented and evaluated.

Milestone 2/3 Preliminary Results

1. Unified Multi-Skill Imitation Learning

Recovery Scenario	**A**	**B**	**C**	**D**
Unified Transformer	**14/19**	**3/4**	**3/19**	**0/17**
Success Rate	73.7%	75.0%	15.8%	0.0%

A/B: recovery from disturbed grasps; C: recatching after contact loss.

Main Achievement

A single Transformer policy successfully performs three different recovery behaviors without an explicit skill selector.

2. Precision Recovery (D)

- A dedicated Precision model achieved **8/8** successes on a small training set.
- On unseen Precision cases, success remained limited to **2/21**.
- Precision recovery has not yet been integrated successfully into the unified policy.

Key Takeaway

Unified multi-skill recovery is feasible, but improving reliability and generalization remains the main challenge.

Milestone 3/3 Challenges and Next Steps

Current Challenge

Imitation learning has shown promising results on individual recovery skills, but generalization and multi-skill performance remain limited.

1. **Improve Supervised Fine-Tuning (SFT).**
   Improve recovery performance on unseen situations and develop a unified policy that can reliably perform different recovery skills.

2. **Reinforcement Learning (RL).**
   Further optimize the learned policy through interaction with the simulation environment, using physical recovery success as the objective.

Final Goal

A unified policy that can recover from different impact-induced grasp failures while preserving stable grasps.